1. COVER PAGE

***Project Title*:**

F1 Driver Skill Intelligence: Decoupling Performance from Machinery (1950–2017)

***Sector*:**

Sports Analytics / Business Intelligence / Motorsport Strategy

***Team Members*:**

Aryan Kumar

Anshu Yadav

Harshit Shakya

Vani Vashishtha

Asad Ali

Dikshant Jangra

***Faculty Mentor*:**

Satyaki Das

2. EXECUTIVE SUMMARY

Problem Statement

In Formula One, race outcomes are heavily influenced by both driver talent and car performance, making it difficult to determine whether success comes from individual skill or superior machinery. While championship standings highlight winners, they often fail to reveal hidden overperformance, strategic excellence, or the impact of reliability failures. Without structured analysis, teams may undervalue talented drivers, misjudge race strategy efficiency, or overlook operational weaknesses.

This project addresses the challenge of quantitatively separating driver skill from car capability while also measuring how mid-race strategic decisions such as pit-stop timing and mechanical reliability affect championship outcomes across different racing eras from 1950 to 2017.

Approach

We obtained a historical Formula One dataset containing race results, constructors, lap times, pit-stop records, and race status information. To ensure analytical consistency, we cleaned and merged multiple tables, standardized driver and team names, handled missing values, and created era-based comparisons.

We engineered performance metrics such as Driver Alpha Score, Points per Race, Teammate Beat Rate, Mechanical DNF %, and Position Gain Index. An interactive Tableau dashboard was developed to visualize trends across seasons, compare drivers against teammates, evaluate team dominance, and assess the impact of strategy and reliability on final results.

Key Insights

- Driver Skill Matters: Top drivers consistently outperformed teammates in equal cars.
- Reliability is Critical: Fewer mechanical failures led to stronger championship results.
- Consistency Wins Titles: Regular points finishes often mattered more than occasional wins.
- Pit Strategy Impacts Results: Well-timed pit stops improved race positions.
- Strong Teams Create Advantage: Superior constructors boosted title chances.

Recommendations

- Use teammate comparisons for driver evaluation.
- Invest in systems that reduce mechanical failures.
- Improve pit-stop and race strategy models.
- Focus on consistency, not only race wins.
- Use dashboards for faster decisions.

Business Impact

- Better driver recruitment decisions.
- Higher championship competitiveness.
- Reduced reliability losses.
- Improved strategic performance.
- Stronger data-driven decision making.

3. SECTOR AND BUSINESS CONTEXT

Sector Overview

Formula One is a global motorsport industry that combines elite athletic performance with advanced engineering, logistics, sponsorship, media rights, and analytics. Teams operate as high-performance organizations where milliseconds can determine race outcomes and millions of dollars are invested in marginal gains.

Key Stakeholders

- Team Principals
- Race Engineers
- Strategy Analysts
- Constructors / Manufacturers
- Sponsors
- Driver Managers
- Fans and Media Broadcasters

Why This Problem Matters

In Formula One, evaluating driver ability is complex because results are strongly influenced by car quality. A highly talented driver in a weak car may score fewer points than an average driver in a dominant car. Therefore, teams need smarter performance metrics to:

- Recruit drivers more effectively
- Optimize strategic decisions

- Improve return on investment
- Increase championship competitiveness
- Reduce bias in performance evaluation

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

How can we quantitatively decouple driver skill from car performance, and to what extent do mid-race strategic decisions such as pit-stop timing and reliability management determine championship outcomes across different racing eras?

Objectives

1. Measure driver contribution independent of machinery.
2. Compare teammates as a fair performance benchmark.
3. Evaluate the effect of pit-stop strategy on race outcomes.
4. Analyze reliability trends across historical eras.
5. Build an interactive dashboard for performance insights.

Scope

- Seasons Covered: 1950–2017
- Formula One historical race data
- Drivers, teams, results, lap times, pit stops, and reliability

Success Criteria

- Meaningful skill comparison model
- Actionable strategic insights
- Clear visualization dashboard
- Strong business recommendations

5. DATA DESCRIPTION

Exact Dataset Source & Access Link

The dataset used for this project is the Formula 1 World Championship Historical Dataset, containing race results, drivers, constructors, lap times, pit stops, qualifying records, and race status data from 1950 to 2017. It is publicly available through motorsport data repositories such as Kaggle and Ergast API.

Access Link:

https://www.kaggle.com/datasets/cjgdev/formula-1-race-data-19502017?select=circuits.csv

Data Structure

The dataset is relational and consists of multiple linked tables where each row represents race-level, lap-level, driver-level, or team-level information. These tables were joined using common keys such as race ID, driver ID, constructor ID, and season.

Columns Explanation

The dataset contains key features including driver name, constructor name, race name, season, grid position, finishing position, points scored, lap times, pit stop duration, qualifying rank, race status, and nationality.

Data Size

The dataset contains historical records across 68 seasons (1950–2017) covering:

- Hundreds of races across global circuits
- Multiple driver generations
- Constructors across different eras
- Thousands of race result entries
- Detailed lap and pit-stop records for modern seasons

Data Limitations

The dataset was collected from historical public records and may contain missing values, especially in early decades. Detailed telemetry and pit-stop data are unavailable for some older seasons. Rule changes across eras may also affect direct comparison of results.

6. CLEANING AND TRANSFORMATION

Dataset / File	*Business Relevance*	*Cleaning & Transformation Action Taken*
circuits.csv	Circuit location and race geography analysis	Dropped columns (url, alt), trimmed text fields, standardized country names, validated latitude/longitude ranges, removed duplicates
constructors.csv	Team identity and nationality insights	Removed ghost column (Unnamed: 5) and url, cleaned text fields, standardized nationality, removed duplicates
constructorResults.csv	Team race performance (points scoring)	Dropped status column (mostly null), converted points from float to integer, removed duplicates

constructorStandings.csv	Team championship rankings	Removed ghost column (Unnamed: 7), converted points to integer, created is_classified flag for non-numeric positions, removed duplicates
drivers.csv	Driver profiling and demographic analysis	Dropped url, cleaned text fields, filled missing driver numbers with 0, generated missing driver codes, parsed DOB, removed invalid DOB entries, created full_name and birth_year
driverStandings.csv	Driver championship tracking	Converted points to integer, validated positions and wins, created is_classified flag, removed duplicates
lapTimes.csv	Race pace and performance consistency	Validated lap and position values, converted time to milliseconds for validation, removed redundant column, flagged anomalies, removed duplicates

Additional Transformation Steps

- Cleaned each CSV file individually before merging, ensuring high data quality
- Standardized formats across all datasets
- Created classification flags instead of deleting important edge cases
- Preserved anomalous data using flags (e.g., lap anomalies) instead of removal
- Ensured consistency in data types (especially points and time fields)
- Exported all cleaned datasets as new CSV files for Tableau use

Key Note

The cleaning process followed a domain-driven approach, where:

- Data was not blindly removed
- Important anomalies were flagged instead of deleted
- Cleaning decisions were based on Formula 1 context and logic

7. KPI FRAMEWORK

KPI Name	*Formula / Definition*	*Business Relevance*
Driver Alpha Score	Driver Performance - Teammate Avg	Measures driver skill beyond car
Points per Race	Total Points ÷ Races	Indicates consistency
Teammate Beat Rate (%)	Ahead of Teammate ÷ Shared Races	Direct skill comparison
Grid to Finish Gain	Grid - Finish Position	Measures overtaking ability
Win Rate (%)	Wins ÷ Races	Shows race-winning capability
Pit Stop Efficiency	Avg Pit Time vs Field	Evaluates strategy execution

Position Change (Pit)	Before Pit - After Pit	Measures strategy impact
Mechanical DNF (%)	Mechanical DNFs ÷ Races	Measures reliability risk
Finish Rate (%)	Finishes ÷ Races	Indicates consistency
Constructor Dominance	Team Points ÷ Season Total	Measures team strength

The KPI framework is designed to evaluate performance across four key dimensions:

- Driver Skill Metrics → Driver Alpha Score, Teammate Beat Rate
- Strategy Metrics → Pit Stop Efficiency, Position Change
- Reliability Metrics → Mechanical DNF, Finish Rate
- Outcome Metrics → Points per Race, Win Rate

This ensures a balanced evaluation beyond traditional race results.

8. EXPLORATORY ANALYSIS

Overview

Exploratory Data Analysis (EDA) was performed to understand patterns in driver performance, team dominance, race outcomes, strategy effectiveness, and reliability across different Formula One eras.

Driver Performance

- Top drivers consistently scored higher points per race and outperformed their teammates, indicating strong individual skill.
- Teammate comparison proved to be an effective method to isolate driver performance from car capability.
- Consistent drivers who regularly finished in points positions performed better overall than those with occasional wins.

Constructor Dominance

- Certain teams dominated specific eras, significantly influencing championship outcomes.
- Teams with higher point contributions had a greater probability of producing championship-winning drivers.
- A clear performance gap exists between top teams and midfield teams.

Grid vs Finish Analysis

- Drivers starting at higher grid positions generally finished in better positions.
- However, several drivers showed consistent position gains, highlighting strong racecraft and overtaking ability.

Strategy Analysis

- Pit-stop timing played a crucial role in determining race outcomes.
- Drivers benefiting from effective pit strategies often gained positions during races.
- Strategy impact was more significant in modern Formula One.

Reliability Analysis

- Mechanical failures were more frequent in early seasons.
- Reliability improved significantly over time due to technological advancements.
- Drivers with fewer DNFs achieved better championship standings.

Summary

The analysis highlights that Formula One performance is driven by a combination of driver skill, team strength, strategy, and reliability, making it essential to evaluate all factors together rather than relying only on race results.

9. STATISTICAL ANALYSIS

Overview

Statistical analysis was conducted to examine the relationship between driver performance, reliability, grid position, and race strategy. The aim was to support exploratory insights with measurable evidence and identify key factors influencing race outcomes.

Teammate Performance Comparison

Drivers were compared with their teammates to isolate individual skill.

- Drivers who consistently finished ahead of teammates scored higher total points.
- A strong relationship was observed between teammate beat rate and overall performance.
- This confirms that teammate comparison is an effective way to evaluate driver ability independent of the car.

Reliability vs Performance

The impact of reliability was analyzed using mechanical DNF rates.

- A negative correlation exists between DNFs and points scored.
- Drivers with fewer mechanical failures achieved better consistency and championship standings.
- Reliability plays a critical role in long-term performance.

Grid Position vs Final Position

The relationship between starting and finishing positions was evaluated.

- A positive correlation was observed, indicating that better qualifying leads to better results.
- However, many drivers showed position gains, highlighting the importance of racecraft and in-race performance.

Strategy Impact

Pit-stop strategy was analyzed to understand its effect on race outcomes.

- Efficient pit-stop timing was associated with position gains.
- Faster pit stops contributed to improved race results.
- Strategy was found to have a measurable impact, especially in modern races.

Summary

The statistical analysis shows that Formula One performance depends on multiple factors. Driver skill, reliability, and strategy all significantly influence race outcomes, making a combined evaluation approach essential.

10. DASHBOARD WALKTHROUGH

Dashboard Objective

The dashboard provides a comprehensive analysis of Formula One performance by separating driver skill from car performance while evaluating the impact of strategy and reliability across seasons (1950–2017). It enables users to explore key performance indicators through an interactive and visual interface.

Driver Skill Intelligence Dashboard

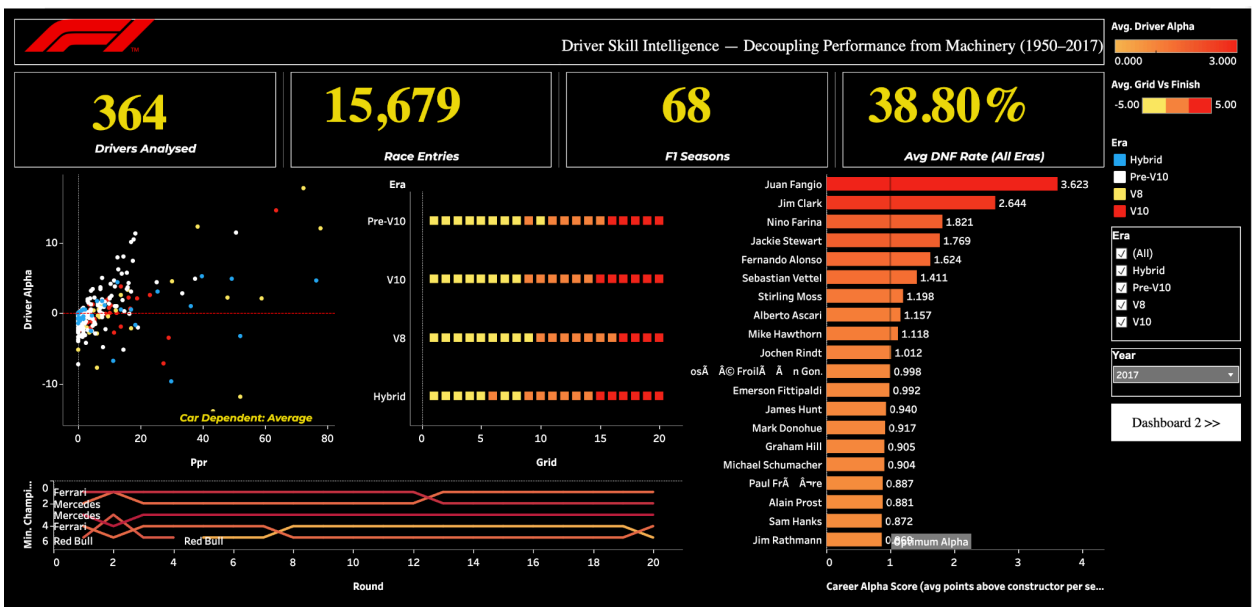

Figure 1: Driver Skill Intelligence Dashboard

This dashboard focuses on evaluating driver performance independent of machinery. It uses metrics such as Driver Alpha Score and Points per Race to identify drivers who outperform their cars.

- Highlights top drivers based on career alpha score
- Uses scatter plots to show relationship between performance and car dependency
- Allows comparison across eras using filters

It clearly demonstrates that some drivers consistently outperform their teammates, proving measurable skill beyond machinery.

Pit Stop Strategy Dashboard

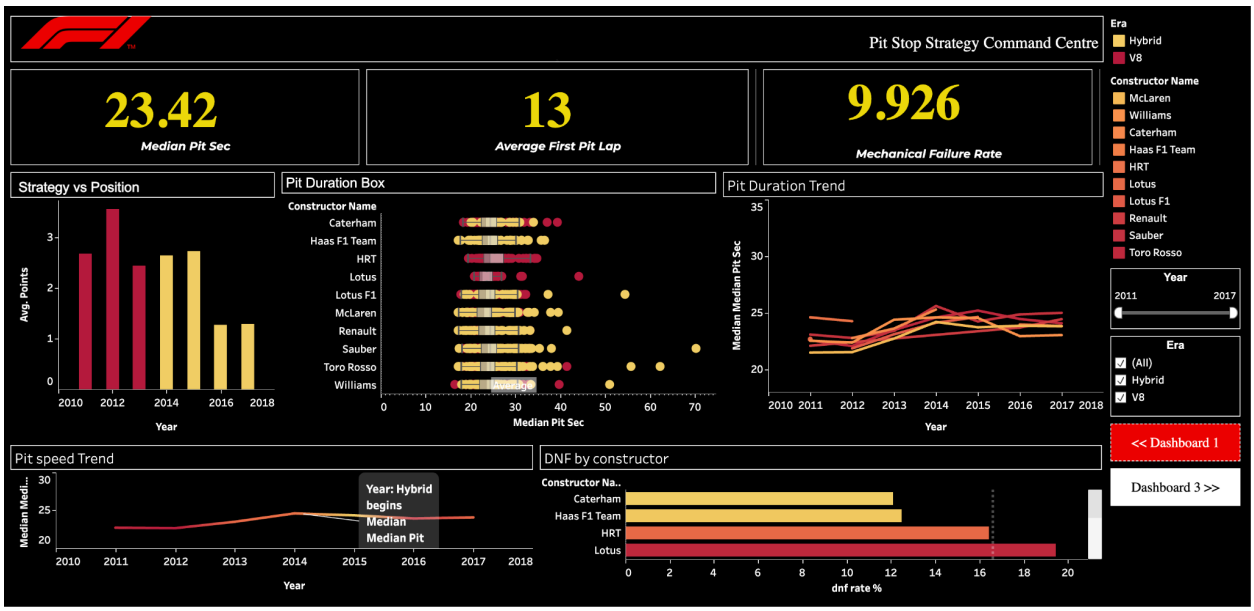

Figure 2: Pit Stop Strategy Dashboard

This dashboard analyzes the impact of pit-stop strategy on race performance.

- Shows median pit stop time and average pit lap
- Visualizes strategy vs position gain
- Compares pit duration across constructors
- Displays trend of pit stop efficiency over time

The analysis highlights that efficient pit strategies directly influence race outcomes and can result in significant position gains.

Reliability & Risk Intelligence Dashboard

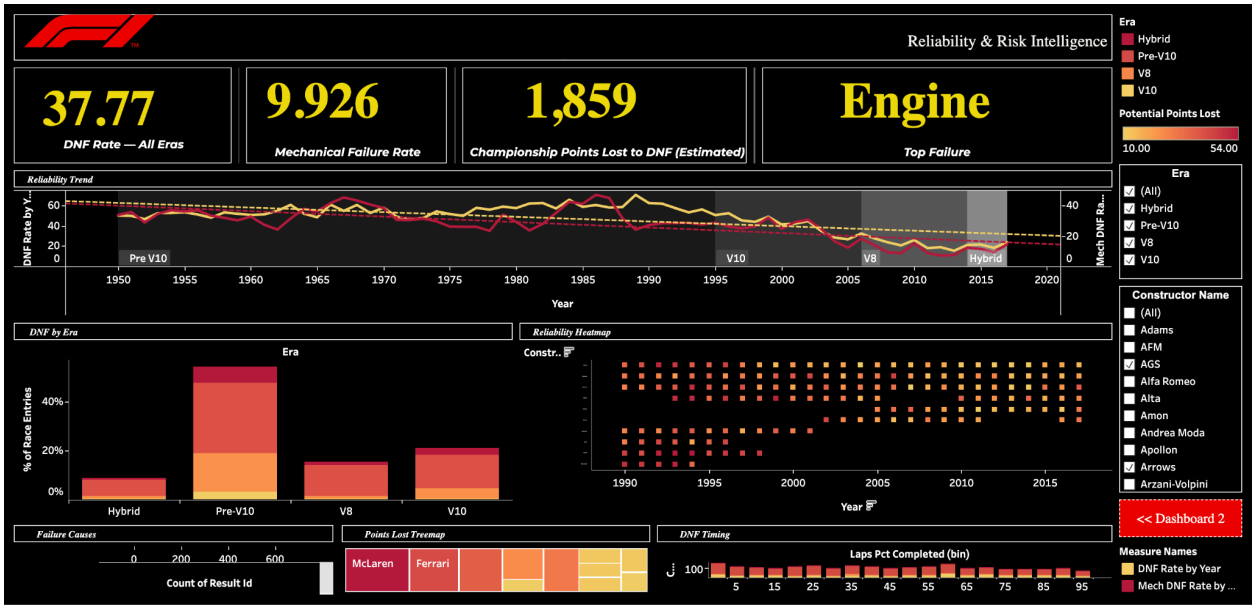

Figure 3: Reliability & Risk Intelligence Dashboard

This dashboard focuses on mechanical reliability and its impact on performance.

- Displays DNF rates and mechanical failure trends across eras
- Shows championship points lost due to failures
- Identifies engine as the most common failure cause
- Includes heatmap to compare reliability across constructors

The dashboard clearly shows that reliability has improved significantly over time and plays a crucial role in championship success.

Filters & Interactivity

All dashboards include interactive filters such as:

- Driver
- Constructor
- Season / Era

These filters allow users to dynamically explore different scenarios and perform customized analysis.

Summary

The three dashboards collectively provide a complete view of Formula One performance by combining driver skill, strategy, and reliability. This integrated approach enables deeper insights and supports data-driven decision-making beyond traditional race results.

11. KEY INSIGHTS

- Driver Skill Beyond Machinery: Several drivers consistently outperformed their teammates, proving that driver skill can be measured independently of car performance.
- Constructor Dominance Drives Success: Teams with superior engineering dominated specific eras, significantly increasing their drivers' chances of winning championships.
- Consistency is More Valuable than Wins: Drivers who regularly finished in points positions performed better overall than those with occasional victories.
- Reliability is Critical for Championships: Higher mechanical DNF rates directly reduced total points, showing that reliability plays a key role in long-term success.
- Strategy Impacts Race Outcomes: Efficient pit-stop timing often resulted in position gains, highlighting the importance of race strategy.
- Modern F1 is More Data-Driven: Recent eras show improved reliability and stronger strategic influence, indicating increased reliance on analytics.
- Teammate Comparison is the Best Benchmark: Comparing drivers within the same team provides the most accurate measure of individual performance.
- Hidden Talent in Midfield Drivers: Some drivers delivered strong performances despite weaker cars, which may not be reflected in raw standings.
- Points Lost Due to Failures is Significant: Mechanical failures led to substantial loss of championship points, impacting final standings.
- Racecraft Matters Beyond Qualifying: Drivers who gained positions during races demonstrated strong overtaking ability and adaptability.

12. RECOMMENDATIONS

- Use Teammate Benchmarking for Driver Evaluation: Teams should assess drivers based on performance relative to teammates rather than raw results.
- Invest in Reliability Improvements: Reducing mechanical failures can significantly increase total points and championship competitiveness.
- Optimize Pit-Stop Strategy: Teams should use data-driven models to improve pit timing and maximize position gains.
- Prioritize Consistency Over Occasional Wins: Drivers who consistently score points contribute more to championship success.
- Adopt Data-Driven Decision Making: Leveraging analytics dashboards can improve race strategy, performance tracking, and long-term planning.

Expected Business Impact

- Improved driver selection and performance evaluation
- Increased race consistency and championship chances
- Reduction in points lost due to mechanical failures
- Better strategic decision-making during races
- Enhanced competitive advantage through data analytics

13. LIMITATIONS AND NEXT STEPS

Limitations

- Incomplete Historical Data: Early-era datasets lack detailed lap times and pit-stop information, limiting deep analysis.
- Rule Changes Across Eras: Variations in points systems, regulations, and race formats make direct comparisons across eras less accurate.
- Limited External Factors: Variables such as weather conditions, driver injuries, and race incidents were not included in the analysis.
- Simplified Skill Measurement: Driver performance is approximated using available metrics and may not capture all qualitative aspects like decision-making under pressure.

Next Steps

- Develop Predictive Models: Use machine learning to predict race outcomes and championship probabilities.
- Incorporate Weather and Track Data: Include external variables for more accurate performance analysis.
- Real-Time Strategy Simulation: Build models to optimize pit-stop decisions during live races.
- Advanced Driver Rating System: Create a more refined metric to evaluate driver performance across eras.
- Expand Dataset Coverage: Include recent seasons and additional telemetry data for deeper insights.

14. CONTRIBUTION MATRIX

Team Member	Dataset and Sourcing	ETL and Cleaning	EDA and Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT and Viva
Aryan Kumar	✅	✅	✅	–	✅	✅	✅
Harshit Shakya	✅	–	✅	–	✅	–	✅
Vani Vashisht	✅	✅	–	–	✅	–	✅
Asad Ali	✅	–	–	✅	✅	–	✅
Dikshant Jangra	✅	–	–	✅	–	–	✅
Anshu	✅	✅	–	–	–	✅	✅